

DICTIONARY

⇒ It is used to store data in key-value pair format.

→ Mutable

→ Ordered

→ Keys should not be duplicate.

⇒ In dictionary, indexing is done on keys.

Initialization

`d1 = {}`

`d1 = dict()`

`d1 = {'a': 1, 'b': 2}`

`d2 = {'a': 1, 'b': [2, 3, 4], 'c': (1, 2)}`

`print(d2['b'])` ⇒ `[2, 3, 4]`

`print(d2['c'])` ⇒ `(1, 2)`

⇒ Key should be an immutable object ⇒ list, set, dict X

• `d1 = {1: "A", 0: "x", 1.0: "AB", True: "Y", False: "W"}`

`print(d1)`

`{1: "A", 0: "x", 1.0: "AB", True: "Y", False: "W"}`

`1.0 == 1`

`True`

`True == 1`

`True`

• `d1 = {'a': 1, 'b': 2}`

for item in d1:

`print(item)`

O/P → `'a'`
`'b'` } key pairs only

• `d1 = dict(a=1, b=2, c=3, d=[1, 2, 3])`

`print(d)`

O/P

`{'a': 1, 'b': 2, 'c': 3, 'd': [1, 2, 3]}`

⇒ Dict Initialization
⇒ Accessing Elements
⇒ Deleting Elements

`{'a': 1, 'b': 2, 'c': 3, 'd': [1, 2, 3]}`

• d2 = dict([('a', 1), ('b', 2), ('c', 3)])

print(d2)

→ {'a': 1, 'b': 2, 'c': 3}

Que. d1 = {'a': 1, 'b': 2, 'c': 3}

d2 = d1

d3 = dict(d1) × (other address)

d4 = {'a': 1, 'b': 2, 'c': 3} × (other address)

d5 = d1.copy() × (other address)

d1['a'] = 7

Which dictionary will affect by changing value in d1.

d1 = {'a': [1, 2], 'b': (7, 8, 9), 'c': {1, 2, 3}, 'd': {'a': 1, 'b': 2, 'c': 3}}

list tuple set dict

Values:

→ Values can be of any datatype

→ Values can be duplicate

→ Values can be modified. # [d1['a'] = 9]

Keys:

→ Keys cannot be mutable objects (list, set, dict)

→ Keys should be only immutable objects

→ Keys cannot be duplicate.

→ Keys can be added or removed but cannot be modified

★ dict.fromkeys (iterable, default)

↓ optional (None)


```
student = ["Anuj", "Anmol", "Patel", "Pankti"]  
→ d1 = dict.fromkeys(student)  
print(d1)
```

o/p → {"Anuj": None, "Anmol": None, "Patel": None, "Pankti": None}

```
→ d1 = dict.fromkeys(student, 0)  
print(d1)
```

o/p → {"Anuj": 0, "Anmol": 0, "Patel": 0, "Pankti": 0}

```
* d3 = {"Anmol": 100, "Pankti": 101, "Chintan": 102}
```

→ d3.keys() ⇒ It will give all the keys in the dictionary

→ d3.values() ⇒ It will give all the values in the dictionary

→ d3.items() ⇒ It will give all key-value pairs in the dict.

```
a = d3.keys()  
print(type(a))
```

→ <class, dict-keys>

```
for item in d3.keys():  
    print(item, end=" ")
```

→ "Anmol", "Pankti", "Chintan"

```
b = d3.values()
```

```
print(type(b))
```

→ <class, dict-values>

```
for i in d3.values():  
    print(i)
```

→ 100

101

102

```
c = d3.items()
```

```
print(type(c))
```

→ <class, dict-items>

```
for i, v in d3.items():  
    print(i, v)
```

→ "Anmol" 100

"Pankti" 101

"Chintan" 102

Accessing the element

d1 = {'a': 1, 'b': 2, 'c': 3}

print(d1['a']) → 1

print(d1['b']) → 2

print(d1['e']) → Error

print(d1.get('a')) → 1

print(d1.get('c')) → 3

print(d1.get('e')) → None

Updating the dictionary

d1 = {'a': 1, 'b': 2, 'c': 3}

d1['a'] = 10

d1.update(a=12, b=30) # d1 = {'a': 12, 'b': 30, 'c': 3}

d2 = {'e': 70, 'f': 90}

d1.update(d2)

print(d1) → d1 = {'a': 12, 'b': 30, 'c': 3, 'e': 70, 'f': 90}

Deletion in dictionary

d1 = {'a': 'GATE', 'b': 2, 'c': 7}

→ del d1['b'] # It will delete the key-value pair where key = 'b'

print(d1) → {'a': 'GATE', 'c': 7}

• del d1 # It will delete complete dictionary
print(d1) → Error

• d1.clear() # It will delete all key-value pairs of dictionary
print(d1) → {}

• a = d1.pop()
print(d1) → {'a': 'GATE', 'b': 2}
print(a) → 7

• a = d1.pop('b')
print(d1) → {'a': 'GATE', 'c': 7}
print(a) → 2